Homework 6

1. Frobenius norm

   (a) Let S be the space of all real $n \times m$ matrices. Define

   $$\langle A, B\rangle_F = \operatorname{tr}\left(A^\top B\right) \ \forall A, B \in S.$$

   Prove that $\langle ., , \rangle_F$ is an inner product on S. Furthermore, show that the induced norm is indeed the Frobenius norm.

   Sol: Since $A^\top \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times m}$, $A^\top B \in \mathbb{R}^{m\times m}$, and

   $$\langle A, B\rangle_F = \operatorname{tr}(A^\top B) = \sum_{j=1}^{m} (A^\top B)_{jj} = \sum_{j=1}^{m}\sum_{i=1}^{n} (A^\top)_{ji}\, b_{ij} = \sum_{i=1}^{n}\sum_{j=1}^{m} a_{ij} b_{ij}.$$

   We check the inner product axioms using this formula. Let $A, B, C \in S$ and $\alpha, \beta \in \mathbb{R}$. It follows that

   $$\langle A, B\rangle_F = \sum_{i,j} a_{ij} b_{ij} = \sum_{i,j} b_{ij} a_{ij} = \langle B, A\rangle_F.$$

   Now, let $\alpha, \beta \in \mathbb{R}$. Then,

   $$\langle \alpha A + \beta B, C\rangle_F = \sum_{i,j} (\alpha a_{ij} + \beta b_{ij}) c_{ij} = \alpha \sum_{i,j} a_{ij} c_{ij} + \beta \sum_{i,j} b_{ij} c_{ij} = \alpha \langle A, C\rangle_F + \beta \langle B, C\rangle_F,$$

   so the operation is linear. Finally, consider

   $$\langle A, A\rangle_F = \sum_{i,j} a_{ij}^2 \geq 0.$$

   This is a sum of squares, so it can only be 0 if all $a_{ij} = 0$ and is strictly positive otherwise. In other words, the operation is positive definite.

   Combined, these properties make $\langle \cdot, \cdot \rangle_F$ an inner product. The induced norm is

   $$\sqrt{\langle A, A\rangle_F} = \left(\sum_{i=1}^{n}\sum_{j=1}^{m} a_{ij}^2\right)^{\frac{1}{2}} = \|A\|_F.$$

   (b) Prove the cyclic property of the trace:

   $$\operatorname{tr}(ABC) = \operatorname{tr}(BCA) = \operatorname{tr}(CAB) \tag{1}$$

   for all $A \in \mathbb{R}^{n\times m}$, $B \in \mathbb{R}^{m\times l}$, $C \in \mathbb{R}^{l\times n}$.

   Sol: Let $A \in \mathbb{R}^{n\times m}$, $B \in \mathbb{R}^{m\times l}$, $C \in \mathbb{R}^{l\times n}$. First,

   $$(AB)_{ij} = a_{i,:} b_{:,j} = \sum_{k=1}^{m} a_{ik} b_{kj}.$$

   Let $AB = P_1$ and notice that

   $$(P_1 C)_{ih} = \sum_{x=1}^{l} p_{1ix} c_{xh} = \sum_{x=1}^{l}\sum_{k=1}^{m} a_{ik} b_{kx} c_{xh},$$

meaning

$$\text{tr}(ABC) = \sum_{i=1}^{n}\sum_{x=1}^{l}\sum_{k=1}^{m} a_{ik}b_{kx}c_{xi}.$$

Following similar logic, we have

$$(BCA)_{ij} = \sum_{k=1}^{n}\sum_{x=1}^{l} b_{ix}c_{xk}a_{kj},$$

and

$$\text{tr}(BCA) = \sum_{i=1}^{m}\sum_{x=1}^{l}\sum_{k=1}^{n} b_{ix}c_{xk}a_{ki}.$$

Since multiplication is commutative and i, x, and k are dummy variables,

$$\begin{aligned}\text{tr}(BCA) &= \sum_{i=1}^{m}\sum_{x=1}^{l}\sum_{k=1}^{n} b_{ix}c_{xk}a_{ki} \\ &= \sum_{i=1}^{n}\sum_{x=1}^{l}\sum_{k=1}^{m} a_{ik}b_{kx}c_{xi} = \text{tr}(ABC).\end{aligned}$$

Finally,

$$\text{tr}(CAB) = \sum_{x=1}^{l}\sum_{k=1}^{m}\sum_{i=1}^{n} c_{xi}a_{ik}b_{kx},$$

which again by commutativity of multiplication provides

$$\text{tr}(CAB) = \text{tr}(BCA) = \text{tr}(ABC).$$

(c) Using the cyclic property, prove that the Frobenius norm is orthogonally invariant. In other words, if $Q_1 \in \mathbb{R}^{n\times n}$ and $Q_2 \in \mathbb{R}^{m\times m}$ are both orthogonal matrices, then

$$\|Q_1AQ_2\|_F = \|A\|_F \quad \forall A \in S.$$

Sol: Let $Q_1 \in \mathbb{R}^{n\times n}$ and $Q_2 \in \mathbb{R}^{m\times m}$ be orthogonal, meaning $Q_1^\top Q_1 = I_n$ and $Q_2Q_2^\top = I_m$, and let $A \in S$. Then $Q_1AQ_2 \in \mathbb{R}^{n\times m}$, so $Q_1AQ_2 \in S$. It follows that,

$$\begin{aligned}\|Q_1AQ_2\|_F^2 &= \text{tr}\left((Q_1AQ_2)^\top(Q_1AQ_2)\right) = \text{tr}\left(Q_2^\top A^\top AQ_2\right) \\ &= \text{tr}\left(Q_2^\top Q_2A^\top A\right) = \text{tr}\left(A^\top A\right) \\ &= \|A\|_F^2 \\ \Rightarrow \|Q_1AQ_2\|_F &= \|A\|_F.\end{aligned}$$

(d) Finally, prove that

$$\|A\|_F^2 = \sum_{i=1}^{d}\sigma_i^2, \quad d = \min\{m, n\} \tag{2}$$

where σ_i are the singular values of A. *Hint:* use the full SVD of A and part (c).

Sol: Let $A = U\Sigma V^\top$ be the SVD of A, where $U \in \mathbb{R}^{n\times n}$ and $V \in \mathbb{R}^{m\times m}$ are orthogonal and $\Sigma \in \mathbb{R}^{n\times m}$ has $\Sigma_{ii} = \sigma_i$ for $i = 1, \dots, d$. Since $U^\top U = I_n$ and $V^\top V = I_m$, we have $\Sigma = U^\top AV$, where $U^\top \in \mathbb{R}^{n\times n}$ and $V \in \mathbb{R}^{m\times m}$ are orthogonal. From earlier, we must have

$$\|A\|_F^2 = \|U^\top AV\|_F^2 = \|\Sigma\|_F^2 = \sum_{i=1}^{n}\sum_{j=1}^{m}\Sigma_{ij}^2 = \sum_{i=1}^{d}\sigma_i^2.$$

2. Uniqueness of reduced QR factorization of a matrix with full column rank.

   (a) Let $\mathcal{U}$ denote the set of all $m\times m$ upper-triangular matrices with strictly positive diagonal entries. Using the results established in Problem 3 of HW5 for lower-triangular matrices, prove the following:

   i. The product of any two matrices in $\mathcal{U}$ is also in $\mathcal{U}$.

   ii. The inverse of any matrix in $\mathcal{U}$ is also in $\mathcal{U}$.

   iii. If $R \in \mathcal{U}$ is orthogonal, then $R = I$.

   Sol: First, let $R_1, R_2 \in \mathcal{U}$. Since $(R_1R_2)^\top = R_2^\top R_1^\top$ is a product of lower-triangular matrices, it is lower-triangular, so R_1R_2 is upper-triangular. For its diagonal,

$$(R_1R_2)_{ii} = \sum_{k=1}^{m}(R_1)_{ik}(R_2)_{ki} = (R_1)_{ii}(R_2)_{ii} > 0,$$

   hence $R_1R_2 \in \mathcal{U}$.

   Next, let $R \in \mathcal{U}$. Since R is upper triangular,

$$\det R = \prod_{i=1}^{m} r_{ii} > 0,$$

   so R invertible. Then $(R^{-1})^\top = (R^\top)^{-1}$ is the inverse of an invertible lower-triangular matrix, hence lower-triangular, so R^{-1} is upper-triangular. Since R and R^{-1} are both upper-triangular,

$$I_{ii} = (RR^{-1})_{ii} = r_{ii}\,(R^{-1})_{ii} \quad \Longrightarrow \quad (R^{-1})_{ii} = \frac{1}{r_{ii}} > 0.$$

   Hence $R^{-1} \in \mathcal{U}$.

   Finally, let $R \in \mathcal{U}$ be orthogonal, so $R^\top = R^{-1}$. We now know R^{-1} is upper-triangular, while $R^\top$ is lower-triangular. R then must be diagonal since it is both upper- and lower-triangular. Then

$$I = R^\top R = \operatorname{diag}(r_{11}^2, \dots, r_{mm}^2),$$

   so $r_{ii}^2 = 1$ for each i. Since $r_{ii} > 0$, we get $r_{ii} = 1$ for all i, and therefore $R = I$.

   (b) Let $A \in \mathbb{R}^{n\times m}$, where $n \geq m$, have full column rank. Prove that the reduced QR factorization $A = QR$ is unique if the diagonal entries of R are strictly positive. *Hint:* assume that $A = Q_1R_1 = Q_2R_2$ are two reduced QR factorizations. Then what can you conclude about $R_2R_1^{-1}$?

   Sol: Suppose $A = Q_1R_1 = Q_2R_2$ are two reduced QR factorizations of A, where $Q_1, Q_2 \in \mathbb{R}^{n\times m}$ have orthonormal columns ($Q_1^\top Q_1 = Q_2^\top Q_2 = I_m$) and $R_1, R_2 \in \mathcal{U}$. From earlier, we have that R_1 is invertible with $R_1^{-1} \in \mathcal{U}$. It follows that

$$T := R_2R_1^{-1} \in \mathcal{U}.$$

Multiplying $Q_1R_1 = Q_2R_2$ on the right by R_1^{-1} gives $Q_1 = Q_2T$. Therefore

$$I_m = Q_1^\top Q_1 = T^\top Q_2^\top Q_2\, T = T^\top T.$$

Since T is square, $T^\top T = I_m$ means $T^\top = T^{-1}$, so T is orthogonal. Since $T \in \mathcal{U}$ orthogonal, $T = I_m$. Hence $R_2 = R_1$, and $Q_1 = Q_2T = Q_2$, so the reduced QR factorization with strictly positive diagonal entries in R is unique.

3. NMF

(a) Implement Lee and Seung's algorithm with $k = 10$, random initializations of W and H (code already provided), for 200 iterations. Note that you should use an alternating update strategy where in each iteration you first update W and then update H by making use of the updated W. At the end of each iteration, evaluate and save $\|R\|_F = \|A - WH\|_F$. Plot the residual norm as a function of the number of iterations. How many iterations seem to be enough for convergence of the algorithm (when the residual visually stops changing significantly).

Sol: In Python, we implement as

```
resid = -1.0*np.ones(iter_max)

W = W0
H = H0

for i in range(iter_max):
  W = W/(W@H@H.T)*(A@H.T)
  H = H/(W.T@W@H)*(W.T@A)
  resid[i] = np.linalg.norm(A-W@H,ord='fro')
```

and get the output

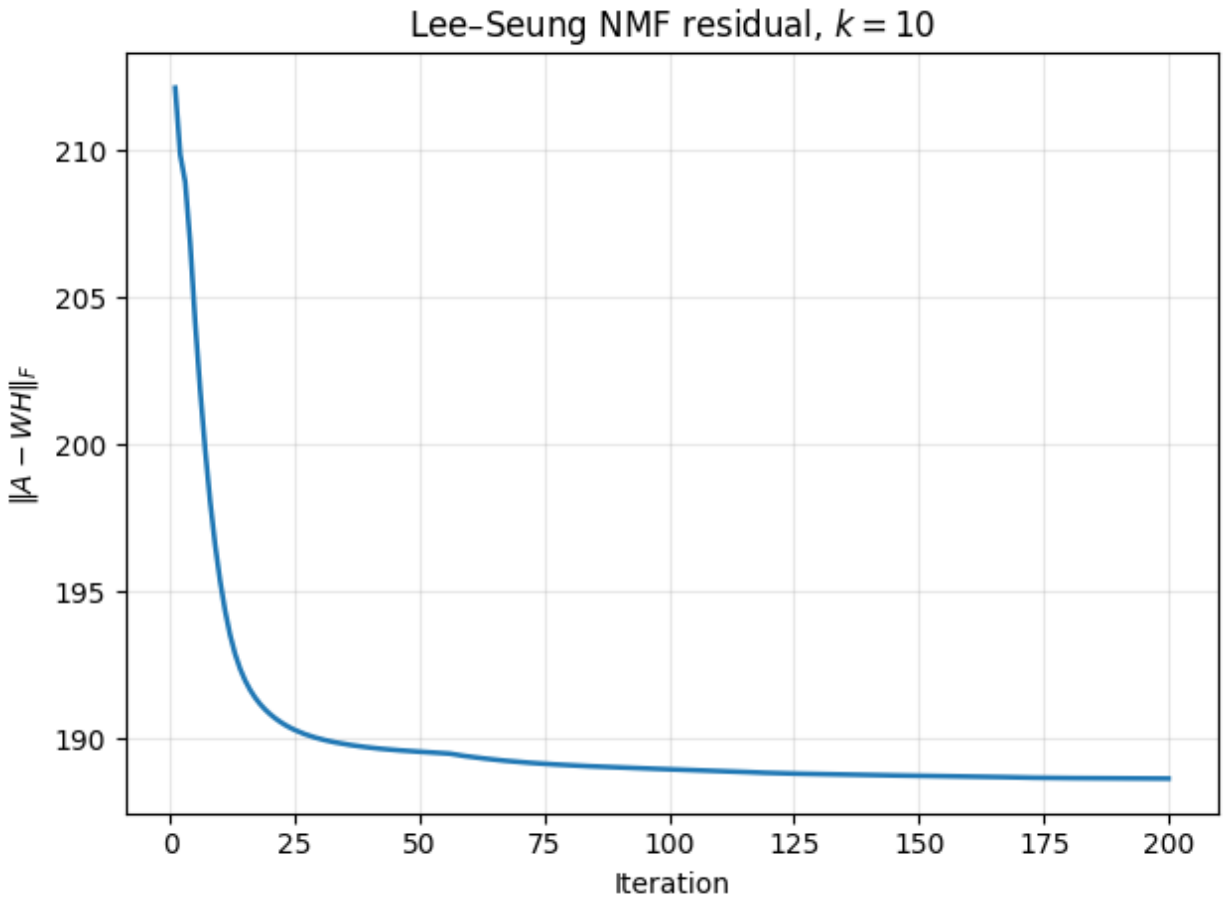


Figure 1: Convergence of Lee and Seung's NMF algorithm with $k = 10$.

Notice from Figure 1 that there is not much change in the error between the NMF approximation and the true data matrix after about 55 iterations.

(b) For each column of the final W, find the top $r = 12$ entries and print out the words corresponding to these entries (code already provided). Based on these words, can you hypothesize what is the commmon theme in each of the k subclasses creating NMF?
Sol:

Table 1: Word Groups

Words	Col 1	Col 2	Col 3	Col 4	Col 5
1	2003	into	atmosphere	and	lyrics
2	links	would	great	the	music
3	page	there	education	for	note
4	gif	next	child	gif	lounge
5	web	people	club	with	including
6	miami	good	calendar	our	lauderdale
7	south	they	each	are	visit
8	the	then	educational	this	groove
9	florida	come	south	you	drummer
10	west	well	creating	all	tracks
11	city	like	community	your	vocals
12	and	when	members	contact	guitarist
Word	**Col 6**	**Col 7**	**Col 8**	**Col 9**	**Col 10**
1	health	south	john	evansville	2003
2	community	miami	martin	2003	show
3	management	florida	during	saturday	great
4	system	lauderdale	old	812	country
5	using	beach	last	indiana	below
6	tested	fort	school	street	music
7	report	west	were	school	radio
8	medical	north	was	history	news
9	given	coral	tell	monday	arts
10	inc	city	complete	sunday	browse
11	children	palm	years	national	these
12	risk	com	volunteers	tuesday	small

Column 9 probably has something to do with southern Indiana, considering Evansville and 812 (the area code there) comes up. Many of the columns (1,5,7) mention Florida or cities in south Florida. Column 5 may have something to do with a band performing in Fort Lauderdale. Similarly, Column 10 mentions music and arts. Column 6 very clearly has medical connotations. Column 3 (and possibly column 8) likely has to do with some community event, maybe hosted by a city government or school. Columns 2 and 4 are hard to interpret since so many of the terms are just prepositions.

(c) Find the best SVD apporximation A_{10} of A. Compute $\|A - A_{10}\|_F$. Compare it with the residuals obtained using NMF.

Sol: In Python, we implement as

```
A_10 = TruncatedSVD(n_components=10).fit_transform(A)
residual = (np.linalg.norm(A-A-10,ord='fro'))/np.linalg.norm(A,ord='fro')
```

and get a much better approximation, with Frobenius error only of 69.3590. When using NMF, the error stabilizes around 190 with $k = 10$, around 140 with $k = 50$, and around 1 with $k = 250$. Of course, the error is going down with larger k, but this is not an apples-to-apples comparison. For $k = 10$, the NMF cannot beat the SVD approximation

in terms of error.